

DWMA: Autonomous Sensorimotor Structure Acquisition, Multi-Schema Retention, and Active Hypothesis Discrimination in an Embodied World Model

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CREDIT AUTHORSHIP CONTRIBUTION STATEMENT

Kunal Lubhana: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing - Original Draft, Writing - Review & Editing, Visualization, Project administration.

DECLARATION OF COMPETING INTEREST

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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DECLARATION OF GENERATIVE AI IN SCIENTIFIC WRITING

During the preparation of this work, the author utilized generative AI and AI-assisted tools (including deep research assistants and large language models) for drafting, restructuring, editorial refinement, LaTeX formatting, and assistance with Python reproduction scripts under continuous human direction and oversight. Following the use of these tools, the author independently audited, executed, validated, and edited all empirical code, mathematical formulations, figure visualizations, and manuscript text. The author assumes full personal responsibility for the factual validity, scientific integrity, and conclusions presented in this publication.

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DATA AND CODE AVAILABILITY

The complete source code, 50-seed deterministic trajectory datasets, configuration files, and turnkey reproduction scripts (`reproduce_all.py`) associated with this study are openly available in the GitHub repository at <https://github.com/Kunallubhana77/DWMA-developmental-world-model-ai> and permanently preserved with open access in the Zenodo research archive at <https://doi.org/10.5281/zenodo.22796871>.